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Target yield based integrated nutrient management improves greengram productivity nutrient uptake and soil biological quality in an Alfisol

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Abstract

Target-yield fertilizer prescriptions are usually validated through crop yield, but their link with rhizosphere biological functioning remains less understood in Alfisol based pulse systems. This short-term field study evaluated whether FYM adjusted STCR-IPNS could improve greengram productivity, nutrient uptake, enzyme activity, microbial response, and PCA derived soil quality index (SQI). Greengram cv. CO 8 was grown in a 12 treatment randomized block design comparing STCR-NPK and FYM adjusted STCR-IPNS at 0.8, 1.0, and 1.2 t ha⁻¹ yield targets with vermicompost, FYM, RDF, RDF + FYM, farmer practice, and control treatments. STCR-IPNS at the 1.2 t ha⁻¹ target produced the highest crop response, with plant height, LAI, and SPAD values of 56.63 cm, 3.98, and 50.76, respectively. The treatment achieved 105.8% of the target yield and recorded higher response ratio, benefit: cost ratio, crude protein content, and crude protein yield than the other nutrient management options. Harvest stage N, P and K uptake peaked at 56.01, 16.96 and 41.43 kg ha⁻¹, respectively. FYM adjusted STCR-IPNS also improved microbial biomass, selected nutrient cycling enzyme activities, and PCA derived SQI. These results indicate that FYM adjusted STCR-IPNS can support greengram productivity while strengthening biological soil quality indicators under the tested Alfisol condition.

Keywords STCR-IPNS, Target yield prescription, FYM nutrient credit, Greengram rhizosphere, Alfisol, Enzyme mediated nutrient cycling, Microbial biomass, PCA derived soil quality index

1 Introduction

High yield fertilizer strategies have improved crop output, but their long-term value now depends on whether they can sustain enzyme-mediated nutrient cycling, microbial regulation, and soil-health function [22, 58]. The problem is no longer fertilizer response alone, but whether yield focused nutrient use can avoid disrupting nutrient balance, rhizosphere biotic processes, and the long-term trade-off between productivity and biological resilience [9,25,51]. Thus, fertilizer prescriptions must be judged not only by



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yield attainment, but also by their capacity to maintain rhizosphere microbial structure and enzyme-regulated soil function [25, 62]. Greengram (*Vigna radiata* L.) is a relevant test crop for this question because it combines protein-rich grain production, biological N inputs, soil organic carbon restoration, and nutrient retention in intensive systems [6, 38]. Integrated nutrient management (INM) partly addresses this challenge by coupling mineral fertilizers with organic inputs that improve biochemical activity, microbial dynamics, and crop productivity [5, 20]. The unresolved step is whether soil-test based nutrient prescriptions can synchronize crop demand with microbial and enzyme-mediated nutrient turnover. The soil test crop response (STCR) approach provides such precision by translating soil fertility status into target-yield fertilizer requirements instead of uniform fertilizer application [36, 42].

For greengram, this precision is agronomically and ecologically important. Fertilizer management must support yield, grain quality, N acquisition, and biologically active soil function at the same time [29, 38]. Recent greengram studies show that STCR integrated fertilization can improve yield, nutrient acquisition, nodulation, and PCA derived soil quality [31, 36, 44]. However, the enzyme microbial pathway behind this response is still not fully resolved. This gap is important because organic-mineral nutrition can sustain plant nutrient supply through microbially mediated nutrient release and enzyme driven N transformation [14]. Therefore, FYM derived nutrient credits in STCR-IPNS should be tested not only as fertilizer substitutions, but also as carbon linked inputs that influence microbial biomass, respiration, enzyme activity, and soil quality index formation [3, 27, 46, 57]. The novelty of this study lies in extending STCR-IPNS validation beyond conventional yield and nutrient uptake assessment. Previous studies on greengram have mainly focused on fertilizer prescription, crop yield, nutrient use efficiency, and basic soil fertility response. However, limited information is available on how FYM-adjusted STCR-IPNS regulates enzyme mediated nutrient cycling, microbial biomass, microbial population dynamics, respiration behaviour, and PCA derived biological soil quality in Alfisol based pulse systems. Therefore, this study provides an integrated crop-rhizosphere evaluation by linking target yield fertilizer prescription with productivity, NPK acquisition, soil enzyme activity, microbial functional response, and soil quality index under greengram cultivation. We hypothesized that FYM-adjusted STCR-IPNS would improve greengram productivity by synchronizing target-yield nutrient supply with NPK uptake, rhizosphere enzyme activity, microbial functional response, and PCA-derived soil quality. This approach makes the study distinct from earlier STCR based greengram studies because it evaluates whether FYM nutrient credits can improve not only target yield achievement, but also the biological functioning of the rhizosphere. Thus, the work contributes a soil-health oriented validation framework for STCR-IPNS in greengram grown on Alfisol. The objectives were to (i) quantify green gram productivity, yield partitioning, grain quality, and NPK acquisition under STCR-NPK, FYM adjusted STCR-IPNS, organic, blanket fertilizer, and farmer-practice benchmarks; (ii) determine the corresponding rhizosphere response in soil fertility, enzyme activity, microbial biomass, microbial population, respiration, and metabolic quotient; and (iii) integrate biological and biochemical indicators through PCA derived SQI to identify the nutrient-management pathway most strongly linked with soil health and crop performance.

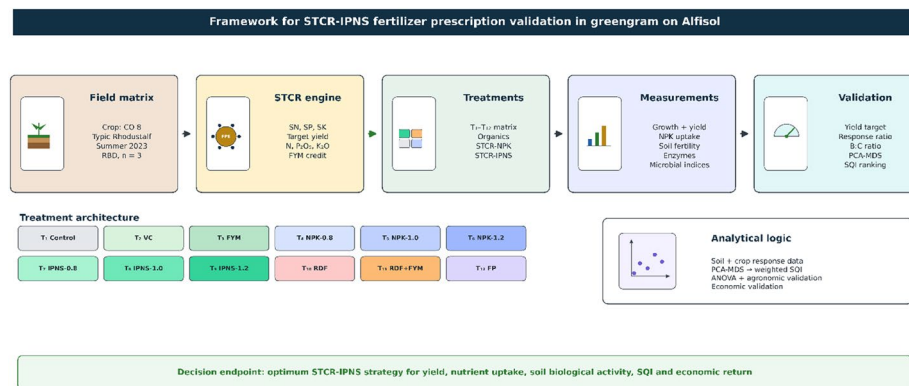


Fig. 1 Conceptual validation framework

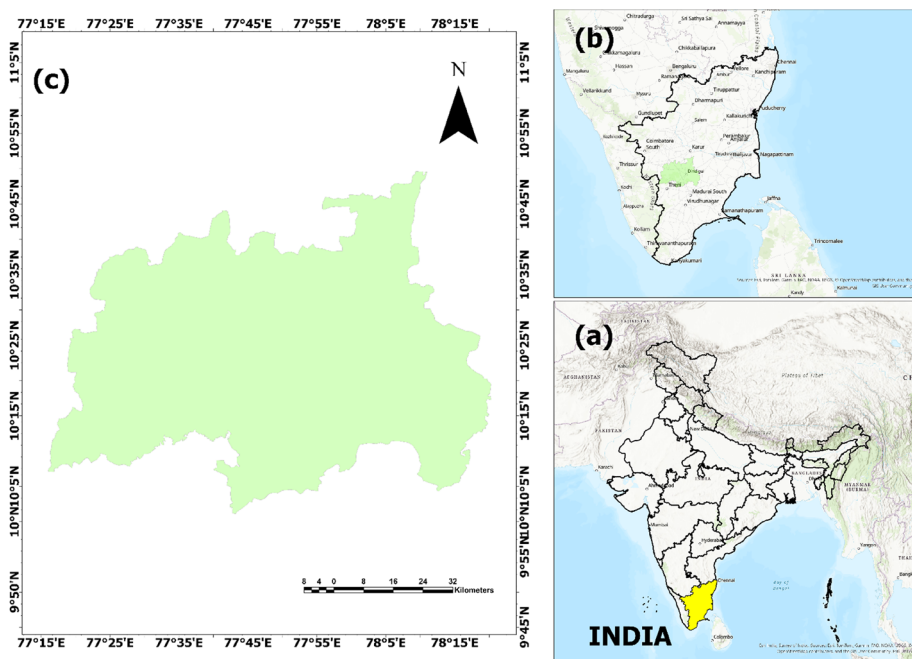


Fig. 2 Location map of the greengram STCR-IPNS validation site. **a** India showing Tamil Nadu, **b** Tamil Nadu showing Dindigul district, and **c** Dindigul district, where the Poolampatty field site in Vagarai block was located for the Alfisol-based greengram STCR-IPNS validation experiment

2 Materials and methods

2.1 Site-soil context and validation framework

The STCR-IPNS validation platform (Fig. 1) was established for green gram (*Vigna radiata* L.) cv. CO 8 at Poolampatty, Vagarai block, Dindigul district, Tamil Nadu, India (10.557° N, 77.569° E; Fig. 2). The experiment was conducted during one green gram growing season and is therefore presented as a short-term field validation of FYM adjusted STCR-IPNS under Alfisol conditions. During crop growth, the site had a warm semi-arid monsoonal environment, with mean maximum and minimum temperatures of 34.15 and 21.5 °C, relative humidity of 71.0–95.0%, and 26.6 mm rainfall over four rainy days during the cropping period (Table 1). These weather conditions are reported to clarify the seasonal environment under which treatment responses were obtained and to support cautious interpretation of field-scale agronomic recommendations.

Table 1 Weather conditions during the greengram cropping period. Source: Agro Climate Research Centre, Tamil Nadu Agricultural University, Coimbatore, India [2]

Month	Tmax range (°C)	Tmax mean (°C)	Tmin range (°C)	Tmin mean (°C)	RH range (%)	RH mean (%)	RF (mm)	RD
Feb-23	26.6–35.4	32.2	18.9–21.6	21.3	73.0–95.0	82.6	16.7	1
Mar-23	28.5–33.5	33.1	16.8–22.0	21.5	74.0–92.0	84.6	0	0
Apr-23	35.5–37.5	34.4	16.4–24.6	21.8	71.0–92.0	81.4	9.9	3
May-23	36.5–38.2	36.9	16.5–23.5	21.4	72.3–94.8	82.9	0	0
Overall range	26.6–38.2	34.15	16.4–24.6	21.5	71.0–95.0	82.9	-	-

Tmax maximum temperature, *Tmin* minimum temperature, *RH* relative humidity, *RF* rainfall, *RD* number of rainy days during the month

Overall values represent the range or mean across the greengram cropping period from February to May 2023

Table 2 Baseline soil fertility and manure composition

Parameter	Value	Unit	Fertility status
A. Baseline soil fertility			
Soil series	Palaviduthi series; Typic Rhodustalf	-	Red, non-calcareous Alfisol
Texture	Sandy loam	-	Light-textured
Bulk density	1.3	Mg m ⁻³	Moderate
Particle density	2.15	Mg m ⁻³	Baseline
Porosity	39.17	%	Moderate pore space
Soil pH	8.02	1:2.5 soil-water	Mildly alkaline
EC	0.08	dS m ⁻¹	Non-saline
CEC	18.6	C mol(p ⁺) kg ⁻¹	Moderate
SOC	0.35	%	Medium
Available N	230	kg ha ⁻¹	Low
Available P	25	kg ha ⁻¹	High
Available K	370	kg ha ⁻¹	High
DTPA-Mn	4.02	mg kg ⁻¹	Adequate
DTPA-Cu	0.48	mg kg ⁻¹	Deficient
DTPA-Fe	1.13	mg kg ⁻¹	Deficient
DTPA-Zn	0.78	mg kg ⁻¹	Deficient
B. Manure composition			
FYM	12.5	t ha ⁻¹	Organic input
Moisture content	25.5	%	Moisture correction
Total N	0.56	%	N credit
Total P	0.34	%	P ₂ O ₅ credit
Total K	0.55	%	K ₂ O credit

EC electrical conductivity, *CEC* cation exchange capacity, *SOC* soil organic carbon, *FYM* farmyard manure, *DTPA* diethylenetriaminepentaacetic acid

Available N, P, and K refer to alkaline KMnO₄-N, Olsen-P, and NH₄OAc-K, respectively. FYM nutrient composition is expressed as N, P₂O₅, and K₂O on a percentage basis, consistent with the nutrient expression convention used for fertilizer prescription. N, P₂O₅, and K₂O credits indicate the manure derived nutrient contributions considered in the STCR-IPNS fertilizer prescription

The experimental soil was a red, non-calcareous sandy loam of the Palaviduthi series, classified as Typic Rhodustalf. Baseline soil showed mild alkalinity (pH 8.02), non-saline status (EC 0.08 dS m⁻¹), moderate CEC (18.6 cmol(p⁺) kg⁻¹), low OC (0.35%), and available N, P, and K of 230, 25, and 370 kg ha⁻¹, respectively (Table 2) [17, 18, 32, 33, 45, 52]. DTPA-extractable Mn, Zn, Cu, and Fe were 4.02, 0.78, 0.48, and 1.13 mg kg⁻¹,

respectively [24]. Bulk density, particle density, and porosity were 1.30 Mg m^{-3} , 2.15 Mg m^{-3} , and 39.17%, respectively.

2.2 Treatment architecture and fertilizer prescription

The experiment included 12 treatments arranged in a RBD with three replications. The treatments included absolute control, organic only inputs, STCR-NPK, FYM adjusted STCR-IPNS, recommended dose of fertilizer, recommended dose of fertilizer + FYM, and farmers practice. Paired STCR-NPK and STCR-IPNS treatments were tested for 0.8, 1.0, and 1.2 t ha^{-1} yield targets. Fertilizer N, P_2O_5 , and K_2O requirements were calculated using greengram-specific STCR prescription equations. In STCR-IPNS treatments, FYM derived N, P_2O_5 , and K_2O credits were deducted from the corresponding inorganic fertilizer requirements. Organic-only, RDE, RDE + FYM, farmer's practice (FP), and control treatments served as non-prescription benchmarks. The treatment structure was designed to compare fertilizer only target yield prescription with FYM-adjusted target yield prescription and commonly used non-prescription nutrient-management practices, thereby aligning the study with agricultural production, soil fertility management and soil-quality evaluation.

The target yield term (T) in the STCR prescription equations was entered in q ha^{-1} , whereas the treatment labels are expressed in t ha^{-1} . Thus, the 0.8, 1.0, and 1.2 t ha^{-1} targets correspond to 8, 10, and 12 q ha^{-1} respectively, for fertilizer-dose calculation. The calculated fertilizer doses were then adjusted using the STCR validation limits. A maintenance dose of 50% of the blanket recommendation was applied where the calculated value was lower than the maintenance level, and a maximum dose of 150% of the blanket recommendation was applied where the calculated value exceeded the upper limit. As the blanket recommendation for greengram was 25:50:25 kg N: P_2O_5 : K_2O ha^{-1} , the maintenance dose was 13:25:13 kg ha^{-1} and the maximum dose was 38:75:38 kg ha^{-1} after rounding. In STCR-IPNS treatments, FYM derived N, P, and K credits were deducted from the calculated inorganic fertilizer requirement before applying these maintenance and maximum dose limits. Therefore, the dose difference between matched STCR-NPK and STCR-IPNS treatments is not constant across yield targets, because some calculated values were adjusted to the maintenance or maximum dose limits. The STCR prescription equations used in this study were derived for greengram cv. CO 8 on Palaviduthi Alfisol using the inductive cum targeted yield approach of Ramamoorthy et al., [35] and the AICRP-STCR methodology. The equations were based on soil-test values, nutrient requirement, soil nutrient contribution, fertilizer nutrient contribution and FYM nutrient contribution.

2.3 Crop management and stage wise sampling

Greengram was established after field preparation at $30 \text{ cm} \times 10 \text{ cm}$ spacing. Nutrient inputs were applied treatment-wise, while non-nutrient agronomic operations were kept uniform across treatments. Crop and rhizosphere observations were recorded at vegetative, flowering, and harvest stages. Plant samples were used for growth, physiological, yield, grain-quality, and NPK uptake measurements, whereas rhizosphere soil samples were analyzed for fertility, (Table 3)

Table 3 STCR-IPNS validation matrix

Strategy	Fertilizer prescription equation						
STCR-NPK	FN = 9.75T – 0.29SN; FP ₂ O ₅ = 12.12T – 2.83SP; FK ₂ O = 8.65T – 0.14SK						
STCR-IPNS	FN = 9.75T – 0.29SN – 0.70ON; FP ₂ O ₅ = 12.12T – 2.83SP – 0.79OP; FK ₂ O = 8.65T – 0.14SK – 0.62OK						
Treatment	Domain	Target yield t ha ⁻¹	N kg ha ⁻¹	P ₂ O ₅ kg ha ⁻¹	K ₂ O kg ha ⁻¹	Organic input t ha ⁻¹	Validation role
T1	Absolute control	-	0	0	0	-	Zero input
T2	Organic alone	-	0	0	0	VC 5.0	Organic baseline
T3	Organic alone	-	0	0	0	FYM 12.5	FYM baseline
T4	STCR-NPK	0.8	13	26	17	-	NPK-0.8
T5	STCR-NPK	1	31	50	35	-	NPK-1.0
T6	STCR-NPK	1.2	38**	75**	38**	-	NPK-1.2
T7	STCR-IPNS	0.8	13	25	13	FYM 12.5	IPNS-0.8
T8	STCR-IPNS	1	13	42	15	FYM 12.5	IPNS-1.0
T9	STCR-IPNS	1.2	26	66	32	FYM 12.5	IPNS-1.2
T10	Blanket RDF	-	25	50	25	-	RDF
T11	Blanket + FYM	-	25	50	25	FYM 12.5	RDF + FYM
T12	FP	-	15	10	10	-	FP

FN, FP₂O₅, and FK₂O denote fertilizer N, P₂O₅, and K₂O, respectively; doses are expressed as kg ha⁻¹

T target yield, SN, SP, and SK, soil-available N, P, and K; ON, OP, and OK, manure-derived N, P, and K credits STCR soil test crop response, IPNS integrated plant nutrition system, FYM farmyard manure, VC vermicompost, RDF recommended dose of fertilizer, FP farmer's practice

Organic input is expressed as t ha⁻¹

* Maintenance dose, equivalent to 50% of RDF; ** maximum dose, equivalent to 150% of RDF. RDF was 25:50:25 kg N: P₂O₅:K₂O ha⁻¹

Target yield values of 0.8, 1.0, and 1.2 t ha⁻¹ were entered in the STCR equations as 8, 10, and 12 q ha⁻¹, respectively enzyme activity, microbial biomass, microbial population, respiration, and metabolic quotient, enabling linkage with PCA derived SQI

2.4 Crop productivity, nutrient acquisition, and grain-quality measurements

Growth, physiological, and nodulation responses were assessed using plant height, leaf number, leaf area index, SPAD value, chlorophyll traits, photosynthetic pigments, and nodule number. Yield formation was quantified using pods plant⁻¹, seeds pod⁻¹, test weight, grain yield, haulm yield, total biomass, and harvest index. Nutrient acquisition was estimated as N, P, and K uptake in grain, haulm, and total biomass. Grain quality was characterized by crude protein, crude fibre, total sugars, and true protein. Crude protein content was calculated from grain nitrogen content using a conversion factor of 6.25 [28]; crude fibre was determined by the acid and alkali digestion method [21], total sugars were determined by the anthrone colorimetric method [59], and true protein was determined by the Folin phenol method [26]. Prescription performance was evaluated using percent achievement, response ratio and benefit: cost ratio. Percent achievement was calculated using Eq. (1), response ratio using Eq. (2), and benefit: cost ratio using Eq. (3):

$$\text{Percent achievement (\%)} = (\text{Observed grain yield} / \text{Targeted grain yield}) \times 100 \quad (1)$$

$$\text{Response ratio (kg kg}^{-1}\text{)} = \frac{(\text{Grain yield in treatment} - \text{Grain yield in control})}{\text{Total fertilizer nutrient dose applied as N} + \text{P}_2\text{O}_5 + \text{K}_2\text{O}} \quad (2)$$

$$\text{Benefit : cost ratio} = \text{Gross return} / \text{Total cost of cultivation} \quad (3)$$

Gross return was calculated from grain and haulm yield using prevailing local market prices during the 2023 crop season. Total cost included seed, fertilizers, FYM or vermicompost, labour, irrigation, plant protection, field preparation and other

treatment-specific input costs. The price and cost assumptions used for economic analysis are provided in Supplementary Table SM3.

2.5 Rhizosphere fertility, enzyme activity, and microbial functional indicators

Rhizosphere fertility and carbon pools were assessed at vegetative, flowering, and harvest stages using available N, P, K, SOC, labile carbon, and water-soluble carbon. Enzyme-mediated nutrient cycling was characterized through urease, alkaline phosphatase, β -glucosidase, nitrate reductase, and dehydrogenase, representing N, P, C, and redox-linked biochemical processes. Urease activity was determined following Tabatabai and Bremner [48], alkaline phosphatase following Tabatabai and Bremner [47], dehydrogenase following Casida et al., [10], β -glucosidase following Eivazi and Tabatabai [12], and Nitrate reductase activity in rhizosphere soil was determined from NO_2^- -N produced during anaerobic incubation [1]. Soil was incubated under waterlogged conditions with KNO_3 as substrate and 2,4-dinitrophenol to inhibit further reduction of nitrite at 25 °C for 24 h, after which the accumulated NO_2^- -N was determined colorimetrically.

Microbial functional response was evaluated using microbial biomass carbon, microbial biomass nitrogen, potentially mineralizable nitrogen, bacteria, fungi, actinomycetes, basal respiration, and metabolic quotient. Microbial biomass carbon was estimated by the chloroform fumigation-extraction method of Vance et al., [50], and microbial biomass nitrogen following Brookes et al., [7]. Potentially mineralizable nitrogen was estimated by short-term anaerobic incubation. Five grams of < 2 mm soil were placed in incubation tubes, flooded with distilled water to exclude entrapped air, sealed, and incubated at 40 °C for 7 days. After incubation, 15 mL of 4 M KCl and 0.2–0.3 g MgO were added, and NH_4 -N released during incubation was determined by steam distillation [55]. Bacterial, fungal, and actinomycete populations were enumerated by the serial-dilution plate-count method [34]. Soil suspensions were serially diluted using sterile distilled-water blanks. For bacteria, 1 mL of the 10^{-6} dilution was plated on nutrient agar and incubated at 30 °C for 24–48 h. Actinomycetes were enumerated on Ken Knight agar and incubated at 30 °C for 7 days. Colony counts were expressed as colony-forming units per gram of soil and basal soil respiration was determined from CO_2 evolution [4]. One hundred grams of soil on an oven-dry-equivalent basis were adjusted to field-moisture conditions and incubated in airtight 1-L flasks containing 8 mL of 0.5 N NaOH as a CO_2 trap at 28 °C for approximately 20 h. After incubation, 5 mL of 3 N BaCl_2 and phenolphthalein indicator were added to the NaOH trap, and the unreacted alkali was titrated with 0.5 N H_2SO_4 . Respiration was expressed as CO_2 evolved per unit soil and incubation time. Metabolic quotient was calculated as basal respiration per unit microbial biomass carbon. These indicators linked STCR-IPNS nutrient prescription with nutrient availability, enzyme activity, microbial population dynamics, respiration balance, and soil-health assessment, with the treatment-wise dataset provided in Table SM2.

2.6 PCA derived soil quality index and treatment discrimination

Before PCA, all soil fertility, enzyme, microbial and crop performance indicators were standardized to remove unit effects. PCA was performed using R software. Principal components with eigenvalues > 1 were retained, and varimax rotation was used to improve loading interpretation. Variables with absolute loading values ≥ 0.70 were considered high-loading indicators. When strongly correlated indicators occurred within

the same component, the variable with higher loading value and stronger biological relevance was retained for the minimum data set. Selected indicators were scored using linear scoring functions and combined using PCA-derived weights to calculate SQI using Eq. (4):

$$SQI = \sum (W_i \times S_i) \quad (4)$$

where W_i is the PCA derived weight and S_i is the standardized indicator score. PCA loadings, selected indicators, weights and SQI values are provided in Table SM2d–SM2e.

2.7 Statistical analysis

Treatment effects were analyzed by analysis of variance appropriate for a randomized block design using the AGRES statistical program (Pascal Intel Software Solutions). Statistical significance was assessed at $P \leq 0.05$, and treatment means were compared using the critical difference (CD) at the 5% probability level, with standard error of difference (SEd) reported where applicable. Count and percentage-based variables were analyzed on their original measurement scales. PCA was used for treatment discrimination and SQI construction, and graphical outputs visualized treatment responses, PCA-derived separation, and integrated indices. PCA, indicator standardization, loading selection, and SQI calculation were performed using R software and Microsoft Excel.

3 Results

3.1 Physiological-nodulation control of yield architecture

Across vegetative, flowering, and harvest stages, STCR-IPNS at the 1.2 t ha^{-1} target produced the strongest canopy response, expressed through higher plant height, leaf number, leaf area index, and SPAD values than equivalent STCR-NPK and non-prescription benchmarks (Fig. SM1; Table SM1a). Chlorophyll traits and photosynthetic pigments tracked the treatment order, with FYM adjusted STCR-IPNS maintaining greater canopy greenness and physiological activity than fertilizer-only and benchmark nutrient regimes (Fig. SM2a, b; Table SM1b). Most growth and physiological traits followed the overall STCR-IPNS advantage. However, chlorophyll a at 45 DAS was significantly higher in T6 than in T9, showing that the physiological superiority of T9 was not uniform for every pigment trait. Nodulation showed the same prescription response, placing FYM adjusted STCR-IPNS above STCR-NPK for physiological-nodulation performance (Fig. SM1e; Fig. SM2c, d). The advantage extended to pods plant^{-1} , seeds pod^{-1} , and test weight, particularly at the higher STCR-IPNS target (Fig. 3a, b; Table SM1c). At matched target-yield levels, STCR-IPNS outperformed equivalent STCR-NPK, showing that FYM adjustment strengthened yield-component expression beyond mineral fertilizer prescription alone (Fig. 3c).

3.2 Target-yield biomass allocation under STCR-IPNS

Grain yield, haulm yield, and total biomass differed clearly among treatments. The 1.2 t ha^{-1} STCR-IPNS treatment recorded the highest grain yield and total biomass (Fig. 4a; Table SM1d). This response indicates that the yield advantage of STCR-IPNS was mainly due to greater biomass production rather than improved harvest partitioning (Fig. 4a). Grain yield, haulm yield, and total biomass differed clearly among treatments, with the 1.2 t ha^{-1} STCR-IPNS treatment recording the highest productivity response (Fig. 4a;

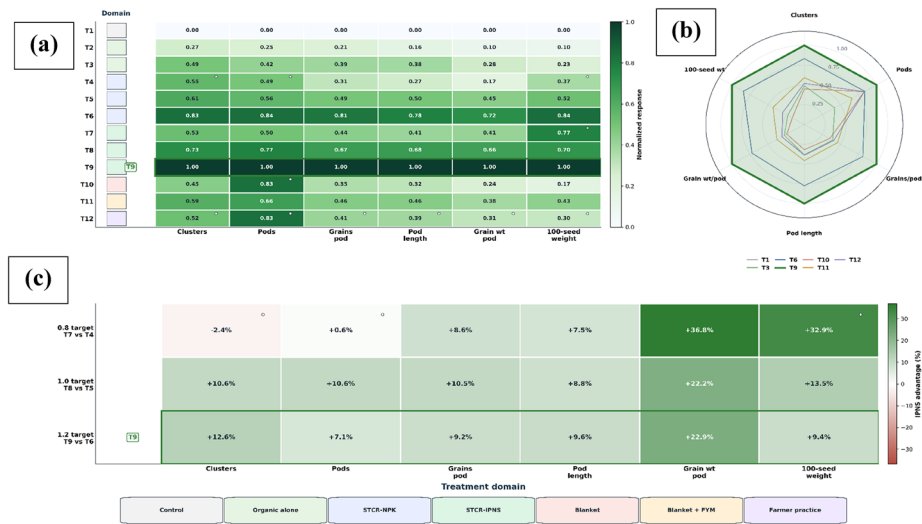


Fig. 3 STCR-IPNS-driven yield-attribute architecture and reproductive sink response. **a** Normalized yield-attribute matrix showing treatment-wise variation in pod and seed traits across nutrient-management domains. **b** Radar based yield-component fingerprint identifying the reproductive sink profile of contrasting nutrient strategies. **c** Matched target-yield comparison showing the percentage advantage of FYM adjusted STCR-IPNS over equivalent STCR-NPK for yield attributes

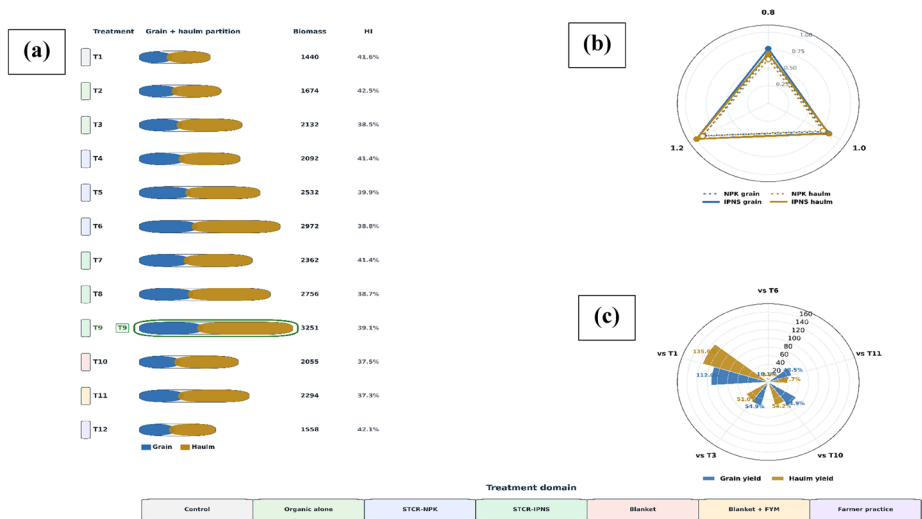


Fig. 4 STCR-IPNS-mediated biomass production, target-yield response, and productivity advantage in Alfisol grown greengram. **a** Biomass partition capsules showing treatment-wise grain yield, haulm yield, total biomass, and harvest index across nutrient-management domains. **b** Target-yield response orbit comparing grain and haulm yield across 0.8, 1.0, and 1.2 t ha⁻¹ STCR-NPK and FYM-adjusted STCR-IPNS prescriptions. **c** Productivity-advantage compass showing the percentage gain of T9 over control, FYM, RDF, RDF + FYM, FP, and equivalent STCR-NPK comparators

Table SM1d). However, this response was mainly due to greater total biomass production rather than a clear improvement in harvest index. T9 recorded the highest productivity advantage across the treatment set (Fig. 4c). Percent achievement, response ratio, and benefit: cost ratio supported the agronomic and economic validity of STCR-IPNS (Fig. SM4; Table SM1e).

Table 4 Productivity-nutrient acquisition coupling

Treatment	Domain	Grain yield	Haulm yield	HI	N uptake	P uptake	K uptake	RR
T1	Control	599	841	41.6	35.98	7.95	24.59	–
T2	Organic alone	712	962	42.5	38.36	9.21	27.89	–
T3	Organic alone	820	1312	38.5	40.75	9.51	31.97	–
T4	STCR-NPK	753	1080	41.1	41.7	9.64	33	2.8
T5	STCR-NPK	1009	1523	39.8	44.69	11.77	35.91	3.5
T6	STCR-NPK	1153	1819	38.8	50.8	14.28	38.67	3.7
T7	STCR-IPNS	811	1385	36.9	45.17	11.73	35.75	3.8
T8	STCR-IPNS	1067	1689	38.7	49.32	14.18	38.64	4
T9	STCR-IPNS	1270	1981	39.1	56.01	16.96	41.43	4.5
T10	RDF	770	1285	37.5	40.05	9.32	30.2	1.7
T11	RDF + FYM	855	1439	37.3	43.51	9.83	34.21	2.6
T12	FP	645	894	41.9	36.34	8.53	26.31	1.3
Mean		872	1351	39.5	43.56	11.08	33.21	3.1
SEd		21.36	26.56	–	0.701	0.25	0.81	–
CD _{0.05}		44.3	55.08	–	1.454	0.51	1.67	–

Grain, haulm yield, and NPK uptake are in kg ha⁻¹;

HI harvest index, RR response ratio

Uptake values represent harvest-stage total uptake

SEd standard error of difference; CD_{0.05}, critical difference at P ≤ 0.05; “–”, not applicable. STCR soil test crop response; IPNS, integrated plant nutrition system, FYM farmyard manure, RDF recommended dose of fertilizer, FP farmer's practice

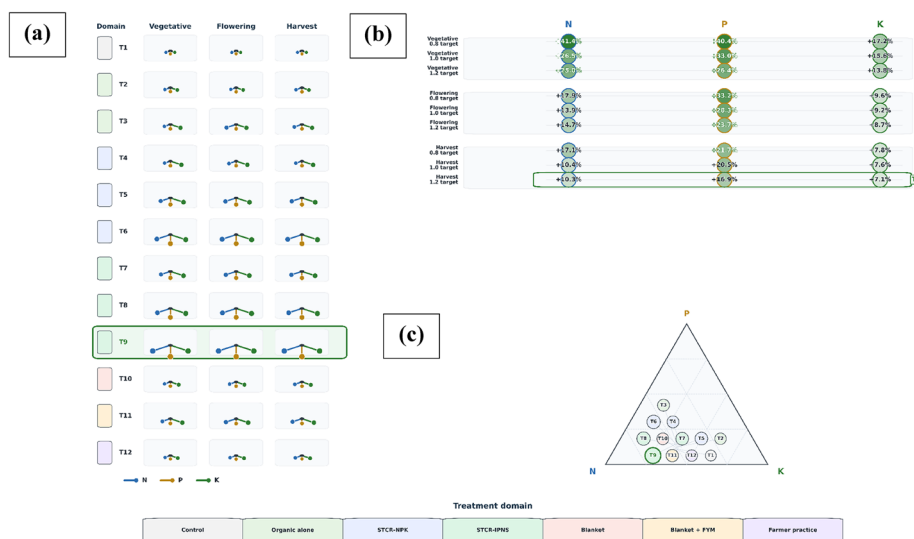


Fig. 5 Nutrient-acquisition stoichiometry and IPNS-derived uptake gain in Alfisol-grown greengram. **a** Uptake glyph matrix summarizing grain, haulm, and total biomass accumulation of N, P, and K across treatment domains. **b** Ternary stoichiometric space showing treatment driven shifts in relative NPK balance under prescription-based, organic, blanket, FP, and control strategies. **c** Matched-target gain bridge quantifying the additional nutrient recovery achieved by FYM-adjusted IPNS over fertilizer-only STCR at equivalent yield targets

3.3 Stoichiometric coupling of nutrient acquisition and yield

N, P, and K uptake across grain, haulm, and total biomass followed the prescription gradient, peaking under STCR-IPNS at 1.2 t ha⁻¹ and declining through lower IPNS targets, matched STCR-NPK, and nonprescription benchmarks (Fig. 5a; Table SM1d). The NPK stoichiometric map shifted treatments from control and organic-only inputs toward STCR-based domains, with STCR-IPNS closest to balanced uptake (Fig. 5b). Productivity nutrient coupling showed larger uptake gains for STCR-IPNS than matched

STCR-NPK, suggesting that organic nutrient credit improved recovery of applied nutrients and yield-linked acquisition efficiency (Fig. 5c; Table 5).

3.4 Nutritional quality modulation under STCR-IPNS

Crude protein and true protein increased most under FYM-adjusted STCR-IPNS, giving the strongest grain-quality response against STCR-NPK and benchmark nutrient strategies (Fig. SM5a; Table SM1e). Fibre and sugar fractions varied with treatment, showing that crude protein enrichment was accompanied by altered carbohydrate partitioning in greengram grain (Fig. SM5a, b). At equivalent target-yield levels, STCR-IPNS maintained a quality advantage over STCR-NPK and produced a more balanced crude protein-sugar quality response (Fig. SM5b, c).

3.5 Enzyme-microbial activation of rhizosphere fertility

Rhizosphere fertility showed a treatment and stage dependent response, with FYM adjusted STCR-IPNS showing clearer improvement in available N, P, and K than fertilizer only STCR-NPK and non-prescription benchmarks (Table SM2a). However, SOC, labile carbon, and water-soluble carbon showed only non-significant treatment related trends across crop stages. Enzyme responses generally followed the treatment pattern, with clearer responses in urease, nitrate reductase, alkaline phosphatase, and dehydrogenase under STCR-IPNS (Table SM2b). β -glucosidase showed treatment related variation, but the difference between T9 and the next-best treatment was not consistently significant across stages. Therefore, β -glucosidase was not used as independent evidence of significantly enhanced carbon turnover. Microbial biomass carbon, microbial biomass nitrogen, and potentially mineralizable nitrogen increased most strongly under STCR-IPNS, alongside higher bacterial, fungal, and actinomycete populations (Table SM2c; Fig. SM3a, b). Basal respiration increased along with microbial biomass, indicating greater microbial activity under integrated nutrient management (Table SM2c; Fig. SM3c, d). However, metabolic quotient did not show a consistent significant decline across stages. Therefore, it was interpreted as a stage dependent respiration indicator rather than as confirmed evidence of improved microbial carbon use efficiency under STCR-IPNS.

3.6 PCA resolved soil health discrimination

PCA captured the integrated variance of rhizosphere fertility, enzyme activity, microbial indicators, and crop-performance traits, with high-loading variables retained as the minimum data set for SQI construction (Table 5; Table SM2d; Fig. 6a, b). The PCA-derived SQI showed clear stage-wise treatment separation, with FYM-adjusted STCR-IPNS, particularly the higher target-yield level, producing the strongest soil-quality response over STCR-NPK and non-prescription benchmarks (Fig. 6c; Table SM2e). PCA ordination distinguished STCR-IPNS from STCR-NPK, organic-only, RDF, RDF + FYM, FP, and control treatments, indicating stronger soil-health differentiation under FYM-adjusted prescription (Fig. 6a, d). Indicator contribution and SQI ranking identified enzyme activity, microbial biomass, respiration balance, and nutrient availability as dominant SQI drivers; overall, STCR-IPNS was the pathway most closely associated with improved soil health and crop performance (Fig. 6b-d; Table SM2d-e).

Table 5 PCA-MDS soil quality index framework

Stage	Retained PCs	Variance explained (%)	PCA weight	MDS indicators	SQL model	SQL range; mean	Peak SQL
Vegetative	PC1, PC2	85.95	8.30	1.00, 0.09	URE, ACP, BG, FUN, PMN, NR, MBC, MBN; ACT, MQ	1(URE + ACP + BG + FUN + PMN + NR + MBN) + 0.09(ACT + MQ)	T9, 9.17
Flowering	PC1, PC2	90.00	5.77	1.00, 0.06	URE, ACT; SR, MQ	1(URE + ACT) + 0.06(SR + MQ)	T9, 2.11
Harvest	PC1	88.71		1	BG, BAC, ACT, NR, MBN	1(BG + BAC + ACT + NR + MBN)	T9, 5.00

PCs with eigenvalue > 1 were retained; MDS variables were selected from high-loading, non-redundant indicators. PCA weights represent proportional variance contribution to SQL
SQL soil quality index, MDS minimum data set, URE urease, ACP alkaline phosphatase, BG, β -glucosidase, FUN fungi, PMN potentially mineralizable N, NR nitrate reductase, MBC microbial biomass carbon, MBN microbial biomass nitrogen, ACT actinomycetes, MQ metabolic quotient, SR soil respiration, BAC bacteria

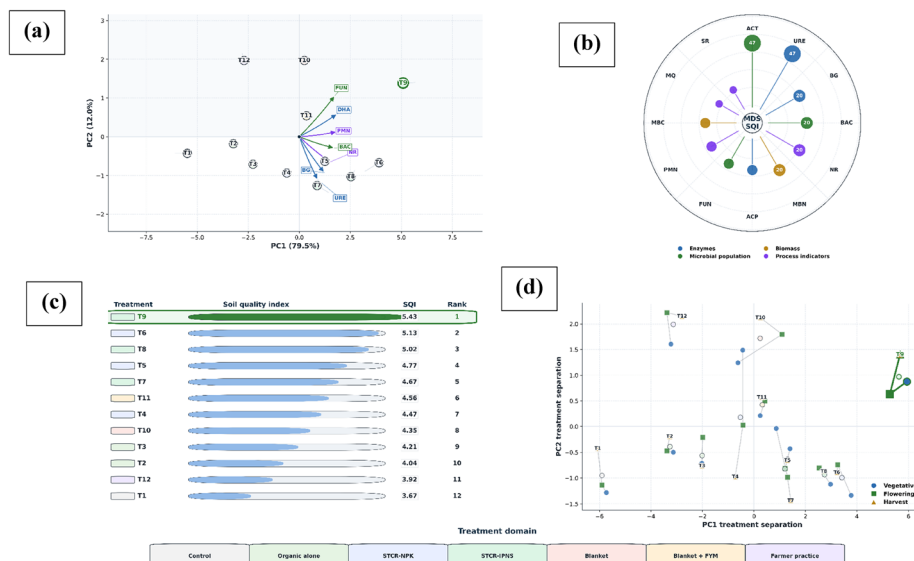


Fig. 6 Multivariate soil-health stratification of greengram rhizosphere under contrasting nutrient-prescription pathways. **a** Ordination biplot separating treatment domains using fertility, carbon-pool, enzyme, microbial, respiration, and crop-performance indicators. **b** Minimum data set contribution profile identifying the dominant variables used for integrated index construction. **c** Soil-quality index ranking across STCR-NPK, FYM-adjusted IPNS, organic, blanket, farmer-practice, and control strategies. **d** Stage-wise separation map showing how rhizosphere functional status diverged across vegetative, flowering, and harvest phases

4 Discussion

FYM adjusted STCR-IPNS improved greengram performance by linking target-yield nutrient prescription with canopy function, nodulation, biomass allocation, and nutrient acquisition [54]. The main contribution of this study is that it connects the agronomic performance of STCR-IPNS with enzyme and microbial indicators of soil biological quality. This is important because target-yield fertilizer prescriptions are usually validated through yield response alone, whereas the present study also evaluates whether the prescription improves rhizosphere functional processes such as nutrient-cycling enzyme activity, microbial biomass formation, microbial population development, and respiration balance. The stronger canopy growth, SPAD response, photosynthetic pigment status, and nodulation indicate that soil-test-based mineral nutrient supply, when combined with FYM-mediated biological activation, created a rhizosphere environment favorable for chlorophyll formation, root activity, N fixation, and reproductive development [43]. It should also be noted that chlorophyll a at 45 DAS was higher under T6 than T9. Therefore, the STCR-IPNS advantage was interpreted as an overall crop performance and soil-biological response rather than as superiority for every individual physiological variable. Similar gains in plant height, leaf area index, nodulation, and yield attributes under integrated nutrient management in greengram have been reported by Kumar et al. [19], while Rath et al. [37] linked nutrient-managed greengram productivity with improved root growth, nodulation, and nutrient uptake; manure substitution has also been associated with higher enzyme activity, bacterial diversity, and crop yield [61]. The yield advantage under FYM adjusted STCR-IPNS was mainly associated with higher grain yield, haulm yield, and total biomass rather than a clear improvement in harvest partitioning. Therefore, the higher productivity of T9 should be interpreted as a biomass driven response supported by better nutrient supply and crop growth, not as a

direct increase in harvest index [11]. The stepwise response from 0.8 to 1.2 t ha⁻¹ targets supports the agronomic relevance of target-yield fertilizer prescription. Higher percent achievement, response ratio, and benefit: cost ratio further indicate the economic advantage of FYM adjusted STCR-IPNS under the tested field condition. Similar gains in

productivity, resource-use efficiency, and nutrient cycling have been reported under integrated nutrient systems [15, 40], and FYM-fertilizer integration has improved soil fertility and biochemical activity [41]. Higher N, P, and K uptake further suggests that STCR-IPNS improved nutrient acquisition as a stoichiometric response, with FYM contributing to sustained nutrient release, root-zone activity, and recovery of applied nutrients. This agrees with reports that manure–fertilizer combinations improve root growth, nutrient uptake, and yield, while organic substitution and integrated fertilization strengthen SOC fractions, enzyme activity, NPK balance, and nutrient recovery [23, 53, 60].

The grain quality response adds another layer to this mechanism. Higher crude protein and true protein, together with changes in fibre and sugar fractions, suggest that STCR-IPNS altered N assimilation and grain quality response in greengram. This is relevant for pulses, where yield improvement should retain protein density and carbohydrate balance [37, 39, 49]. The improvement in crude protein yield was therefore linked not only with higher grain yield, but also with improved N uptake and better physiological performance under integrated nutrient supply.

In the rhizosphere, FYM adjusted STCR-IPNS acted as a biological nutrient-regulation pathway. SOC, labile carbon, and water-soluble carbon likely supplied substrates for microbial biomass, enzyme production, and mineralization. However, the carbon-pool response should be interpreted cautiously because SOC, labile carbon, and water-soluble carbon showed treatment related trends but did not consistently show statistically significant separation across all crop stages. Therefore, these results suggest short-term substrate support from FYM rather than a confirmed strong improvement in carbon pools. The parallel increases in urease, nitrate reductase, alkaline phosphatase, β -glucosidase, dehydrogenase, MBC, MBN, PMN, and microbial populations indicate stronger N transformation, P mobilization, C turnover, oxidative metabolism, and microbial functional capacity. Among these indicators, β -glucosidase was interpreted carefully because the difference between the best-performing treatment and the next-best treatment was not always statistically distinct. Thus, β -glucosidase is discussed as part of the overall enzyme response pattern rather than as an independent conclusive indicator of enhanced C turnover. Increased basal respiration with reduced metabolic quotient further indicates improved microbial carbon-use efficiency rather than inefficient respiratory loss. However, metabolic quotient was not treated as a uniformly significant decline across all stages. Its response is therefore discussed as a stage dependent respiration indicator, not as conclusive evidence of consistently improved microbial carbon use efficiency. Similar organic mineral effects on rhizosphere nutrients, microbial biomass, microbial diversity, soil quality, enzyme activity, SOC fractions, and fertility have been reported under combined amendments, manure substitution, FYM–fertilizer integration, and partial organic substitution [23, 41, 56]. Finally, the PCA-MDS-SQI framework confirmed that STCR-IPNS superiority was not driven by a single variable, but by integrated changes in enzyme activity, microbial biomass, respiration balance, nutrient availability, and crop performance. The clear separation of FYM-adjusted

STCR-IPNS from STCR-NPK and non-prescription benchmarks therefore indicates a more functionally organized rhizosphere, consistent with soil-quality frameworks based on minimum data sets, integrated scoring, and microbial indicators [8, 13, 16, 30]. Although the present study showed clear treatment responses in greengram productivity, nutrient uptake, enzyme activity, microbial indicators, and PCA derived SQI, the results should be interpreted within the limitation of a one-season field experiment. Seasonal variation in rainfall, temperature, residual nutrient dynamics, microbial activity, and crop response may influence the performance of STCR-IPNS under field conditions. Therefore, the findings should be considered as short-term validation evidence rather than a final long-term recommendation. Multi season and multi-location experiments across different Alfisols, soil fertility gradients, and pulse-based systems are required to confirm the stability and wider applicability of FYM adjusted STCR-IPNS.

5 Conclusion

This study shows that FYM adjusted STCR-IPNS can move greengram nutrient management beyond yield based fertilizer prescription. Under the tested Alfisol condition, target-yield fertilization combined with FYM supported crop productivity, NPK uptake, and rhizosphere biological functioning. The strongest response was recorded under the 1.2 t ha^{-1} STCR-IPNS treatment. The novelty of the study is the integration of target yield based STCR-IPNS with biological soil quality indicators in greengram. By combining yield response, nutrient uptake, enzyme activity, microbial indicators and PCA derived SQI, the study provides a broader validation of STCR-IPNS than yield based assessment alone. The study contributes empirical evidence that biological soil quality indicators can strengthen the evaluation of STCR-IPNS beyond conventional yield validation. It also provides a practical framework for integrating soil test-based fertilizer prescription, organic nutrient inputs and rhizosphere functional indicators in pulse-based nutrient management. For field application, however, the reliability of this strategy depends on site-specific soil test calibration, accurate assessment of FYM nutrient composition and realistic consideration of input availability and economic feasibility. Future research should extend this framework across contrasting Alfisols, other soil groups, different pulse crops and multi-season field conditions. Long-term studies should also examine whether the observed improvements in carbon pools, enzyme activity, microbial biomass and PCA derived soil quality are sustained under repeated STCR-IPNS application. By connecting target yield fertilization with enzyme-mediated nutrient cycling and biological soil quality, FYM-adjusted STCR-IPNS offers a forward-looking pathway for improving pulse productivity while maintaining the functional integrity of soil systems.

Abbreviations

STCR	Soil test crop response
IPNS	Integrated plant nutrition system
STCR-IPNS	Soil test crop response–integrated plant nutrition system
STCR-NPK	Soil test crop response with inorganic NPK fertilizers
FYM	Farmyard manure
NPK	Nitrogen: phosphorus and potassium
RDF	Recommended dose of fertilizer
FP	Farmers practice
VC	Vermicompost
RBD	Randomized block design
LAI	Leaf area index
SPAD	Soil plant analysis development chlorophyll meter reading

SCMR	SPAD chlorophyll meter reading
SOC	Soil organic carbon
SQI	Soil quality index
PCA	Principal component analysis
MDS	Minimum data set
MBC	Microbial biomass carbon
MBN	Microbial biomass nitrogen
PMN	Potentially mineralizable nitrogen
URE	Urease
ACP	Alkaline phosphatase
BG	β -glucosidase
NR	Nitrate reductase
DHA	Dehydrogenase activity
BAC	Bacteria
FUN	Fungi
ACT	Actinomycetes
SR	Soil respiration
MQ	Metabolic quotient
HI	Harvest index
RR	Response ratio
B:C ratio	Benefit: cost ratio
EC	Electrical conductivity
CEC	Cation exchange capacity
DTPA	Diethylenetriaminepentaacetic acid
SEd	Standard error of difference
CD	Critical difference

Supplementary Information

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Supplementary Material 1.
Supplementary Material 2.
Supplementary Material 3.
Supplementary Material 4.
Supplementary Material 5.
Supplementary Material 6.
Supplementary Material 7.

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Author contributions

All authors contributed to the study conception and design. Field experimentation, treatment implementation, sampling, laboratory analysis, data compilation, and statistical analysis were performed by Abhirami P., Meeniga Venkateswarlu, S. Maragatham, R. Rajeswari, and D. Balachandar. The first draft of the manuscript was written by Abhirami P., Meeniga Venkateswarlu, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable. This study did not involve human participants, human data, or animals requiring ethical approval.

Plant material collection and use

All methods used in this study complied with relevant institutional, national, and international guidelines and legislation. The plant materials used in this research were cultivated greengram plants *Vigna radiata* L. grown under standard agricultural practices in a collaborating farmer's field under the supervision of the Department of Soil Science, Tamil Nadu Agricultural University, Coimbatore, India. The collection and use of plant materials complied with applicable

local and national regulations. No specific permits or licences were required for this study as the plant materials were cultivated crops and were not collected from protected or wild environments.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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